

Secure Industrial IoT (IIoT) and Smart Grid Anomaly Detector: A Physics-Informed Unsupervised Framework for Edge Deployment

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Abstract—The convergence of operational technology with Internet-connected devices exposes industrial plants and smart grids to attacks that encryption and network segmentation cannot stop, such as an authenticated but compromised sensor that reports plausible yet false measurements. This paper presents a secure Industrial IoT (IIoT) architecture aligned with IEC 62443 zoning, together with a lightweight, unsupervised anomaly detector for smart-grid substation telemetry that complements preventive controls. The detector augments windowed statistics with a physics-consistency residual derived from the three-phase power relation, scores windows with an autoencoder, a local outlier factor model, and the residual itself, fuses the calibrated scores, and suppresses isolated alarms with a 3-of-5 persistence filter. It is trained on normal data only. On a synthetic substation testbed with four injected attack classes (false data injection, replay/freeze, denial-of-service flooding, and ramp drift) and five independent seeds, the fused detector with persistence filtering reaches an F1-score of 0.920 and an ROC-AUC of 0.984. Adding the physics residual raises false-data-injection recall of the same autoencoder from 11% to 100%, indicating that domain invariants matter more than the choice of learning model, whereas score fusion yields no significant gain over an autoencoder with the residual. Limitations, including the synthetic data and a remaining false-alarm burden, are discussed.

Index Terms—Industrial Internet of Things, smart grid, anomaly detection, false data injection, cyber-physical security, autoencoder, IEC 62443, edge computing.

I. INTRODUCTION

Industrial Internet of Things (IIoT) technologies connect sensors, controllers, and analytics platforms to improve the efficiency, flexibility, and observability of manufacturing and energy systems [1], [2]. In the electric sector the same trend appears as the smart grid, which overlays two-way communication and computation on the power system [3]. These cyber-physical systems [4] were historically isolated and engineered for safety and availability rather than security; today their components are long-lived, difficult to patch, and increasingly reachable from enterprise and Internet-facing networks [5].

The consequences of this exposure are well documented. Stuxnet manipulated the logic of industrial controllers while concealing the manipulation from operators [6], and the 2015 attack on Ukrainian distribution utilities left approximately 225,000 customers without power [7]. Surveys of smart-grid security stress that the tight coupling of cyber and physical domains creates threats that conventional information-technology defenses were not designed to handle [8]–[10].

Preventive controls are necessary but not sufficient. Authenticated encryption on the wire (for example TLS 1.3 [11], as profiled for power-system protocols in IEC 62351-3 [12]), zone-and-conduit segmentation [13], and hardened firmware protect channels and boundaries, yet a compromised or faulty field device still authenticates correctly while reporting false values. False data injection (FDI) attacks can even be crafted to pass the residual-based bad-data checks of power-system state estimators when the attacker knows the grid model [14], [15], and the FDI literature is now extensive [16]. A detection layer that reasons about the *content* and *behaviour* of telemetry is therefore an essential complement.

Building such a layer for operational technology (OT) is not straightforward. Labelled attack data are scarce, normal operation is non-stationary (daily load cycles), false alarms are costly for operators, edge hardware is constrained, and alerts must be interpretable. Unsupervised detectors trained only on normal data [17] address the first issue, and small models address the fourth, but generic detectors are blind to attacks that keep every individual measurement inside its normal range while violating the physical relationships *between* measurements.

This paper makes four contributions:

- 1) A defence-in-depth reference architecture for secure IIoT and smart-grid deployments that maps each threat class to a preventive control and a detection cue (Section III).
- 2) A physics-informed, unsupervised detector that combines a power-balance residual, an autoencoder, and a local

outlier factor model, with validation-calibrated score fusion and a persistence filter (Section IV).

- 3) A reproducible synthetic substation testbed with four attack classes and a five-seed evaluation against five baselines, each with and without the physics residual (Sections V and VI).
- 4) A candid analysis of failure modes, alert burden, and threats to validity, including a negative result for score fusion (Sections VI and VII).

Section II reviews related work, Section III presents the threat model and architecture, Section IV describes the detector, Sections V and VI describe and discuss the experiments, Section VII covers limitations and future work, and Section VIII concludes.

II. BACKGROUND AND RELATED WORK

A. IIoT and Smart-Grid Security

Sisinni *et al.* [1] identify energy efficiency, real-time performance, coexistence, interoperability, and security as the main IIoT challenges, and Serror *et al.* [5] observe that IIoT security goals follow chiefly from safety and productivity requirements rather than confidentiality alone. Yan *et al.* [18] and Wang and Lu [10] survey smart-grid communication security; Mo *et al.* [8] argue that the cyber-physical coupling calls for a new “cyber-physical security” discipline, and Sridhar *et al.* [9] evaluate risk jointly over power applications and their supporting cyber infrastructure. A central attack class is FDI: Liu *et al.* [14] show that an attacker who knows the network configuration can craft measurement changes that evade conventional bad-data detection, and Teixeira *et al.* [15] analyse the resulting state-estimator vulnerabilities.

B. Standards and Preventive Controls

NIST SP 800-82 Rev. 3 [19] guides OT security programs. IEC 62443-3-3 [13] defines system security requirements and security levels, and its zone-and-conduit model underlies the architecture in Section III. IEC 62351-3 [12] constrains the use of TLS for TCP/IP-based power-system protocols and, in its 2023 edition, covers both TLS 1.2 and TLS 1.3. These documents govern confidentiality, integrity, and authentication of communications and the hardening of systems; they do not prescribe what to do when authenticated data are nevertheless wrong, which is the gap addressed here.

C. Anomaly Detection for Industrial and Grid Data

Anomaly detection has been surveyed extensively [17]. Classical unsupervised detectors include isolation forest [20], local outlier factor (LOF) [21], and the one-class support vector machine [22]; reconstruction-based approaches use autoencoders [23] or recurrent networks [24]. In industrial settings, Goh *et al.* apply recurrent networks to the SWaT water-treatment testbed [25], [26], Kravchik and Shabtai use one-dimensional convolutional networks on the same data [27], and Chen *et al.* learn and monitor physical invariants of a cyber-physical system [28]. Kitsune shows that ensembles of small autoencoders can perform online network intrusion detection on modest hardware [29], and Zhang *et al.* propose a multi-layer distributed intrusion detection architecture for smart grids [30].

The fog/edge computing paradigm [31] motivates running such detectors near the data source.

D. Gap and Positioning

Most learning-based detectors treat measurements as a generic multivariate series and leave physical laws to be rediscovered from data, while invariant-mining approaches are typically system-specific [36], [37]. We hard-code one cheap, universally valid invariant (three-phase power balance), feed it to standard unsupervised detectors, and quantify its contribution relative to the choice of model in a controlled, reproducible ablation.

III. THREAT MODEL AND SECURE IIOT ARCHITECTURE

A. Threat Model

We consider an adversary who can (i) inject or flood traffic on the OT network, (ii) record and replay messages or freeze a sensor's output, and (iii) compromise a subset of field devices, or their firmware, so that they report manipulated measurements while still passing authentication. We assume the cryptographic primitives are sound, that the edge gateway hosting the detector is not compromised, and that the adversary does not know the detector's parameters and does not adapt to it. The last assumption is restrictive; Section VII discusses attackers who deliberately preserve the physical invariant.

B. Defence-in-Depth Architecture

Fig. 1 shows the proposed architecture, organised into three IEC 62443 zones. In Zone 1, field devices (meters, phasor measurement units, sensors, actuators, PLCs/RTUs) carry a secure element that provides secure boot and signed firmware. In Zone 2, an edge gateway in a demilitarised zone terminates TLS 1.3/DTLS or IEC 62351-3 sessions, attests devices, and hosts the anomaly detector, so telemetry is inspected after authentication and decryption but before it reaches the control centre. In Zone 3, a security information and event management (SIEM) system triages alerts, response playbooks isolate suspicious devices or fall back to a safe state, and a model registry supports threshold recalibration. Placing inference at the edge limits latency and backhaul bandwidth.

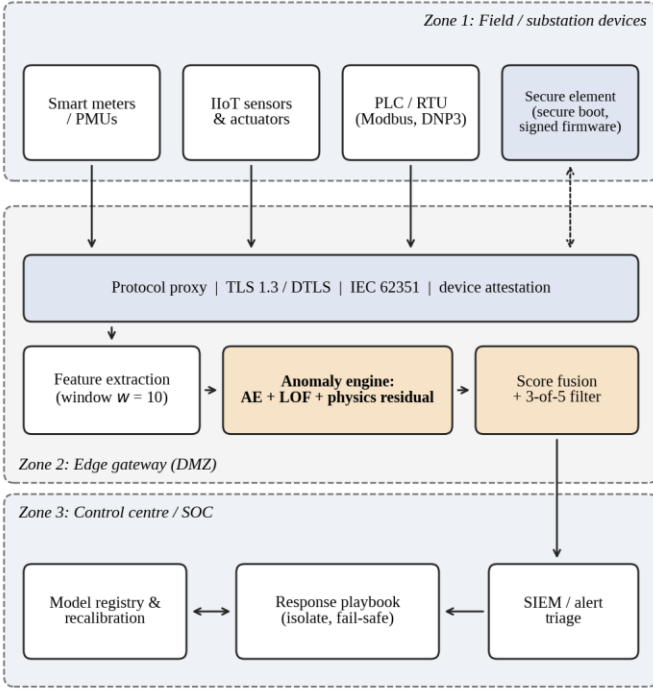


Fig. 1. Proposed defence-in-depth architecture. Preventive controls (shaded blue) protect devices and channels; the anomaly engine and fusion stage (shaded orange) form the detective layer at the edge gateway.

Table I maps the threat classes considered in this paper to a preventive control and to the detection cue that the anomaly detector exploits. The last row is included for completeness but is not evaluated here.

TABLE I
THREAT CLASSES, PREVENTIVE CONTROLS, AND DETECTION CUES

Threat	Preventive control	Detection cue
False data injection / sensor tampering	Secure element, signed firmware, authenticated telemetry	Violated power-balance invariant (r_t)
Replay / frozen sensor	Nonces and sequence counters (TLS 1.3, IEC 62351-3)	Collapse of within-window variance
DoS / traffic flooding	Rate limiting, segmentation (IEC 62443 conduits)	Packet-rate and inter-arrival deviation
Slow drift / calibration attack	Device attestation, periodic recalibration	Growing residual and multivariate distance
Unauthorised control commands	Role-based access, command allow-listing	Out-of-profile protocol behaviour (not evaluated) [38]

IV. ANOMALY DETECTION METHODOLOGY

A. Telemetry Model and Physics-Informed Feature

At each time step t the gateway receives ten channels: line voltage V_t , current I_t , active power P_t , reactive power Q_t , power factor PF_t , frequency f_t , transformer temperature T_t , and three network descriptors (packet rate, mean packet size, and inter-arrival jitter). For a balanced three-phase feeder, active power is tied to line voltage, current, and power factor, so we define the normalised power-balance residual

$$r_t = (P_t - \sqrt{3} V_t I_t PF_t / 1000) / \max(P_t, 1) \quad (1)$$

with V in volts, I in amperes, and P in kilowatts. Under normal operation r_t fluctuates around zero with a small measurement-noise spread; tampering with any one of the four quantities breaks the relation. Together with the ten raw channels this gives an eleven-channel vector z_t . For each window of $w = 10$ consecutive samples (stride 1) we compute the mean and standard deviation of every channel, yielding a 22-dimensional feature vector x_t , which is standardised with training-set statistics. The variant without the residual uses only the ten raw channels (20 features) and serves as the ablation baseline.

B. Detector Branches

1) *Autoencoder (AE)*. A fully connected network with hidden layers of 24, 8, and 24 units (1,518 trainable parameters for 22 inputs) is trained with the Adam optimiser [32] on normal windows only, using early stopping and the scikit-learn implementation [33]. The anomaly score is the mean squared reconstruction error

$$e(x) = (1 / 2d) \|x - g(x)\|^2 \quad (2)$$

where $g(\cdot)$ is the network output and $d = 22$ is the feature dimension.

2) *Local outlier factor (LOF)*. LOF [21] is used in novelty mode with $k = 35$ neighbours, fitted on a random subset of 4,000 training windows so that the stored reference set stays small.

3) *Physics branch*. The absolute window-mean of r_t is used directly as a third score, so that a physically inconsistent window is flagged even when the learned models generalise across it.

C. Score Calibration and Fusion

Raw scores from different models are not comparable. For each branch $m \in \{\text{AE}, \text{LOF}, \text{physics}\}$ we take the logarithm of the score and standardise it robustly with the median and median absolute deviation (MAD) of the same quantity on a held-out *validation* set of normal windows:

$$z_m(x) = (\ln s_m(x) - \text{med}_m) / (1.4826 \cdot \text{MAD}_m) \quad (3)$$

The fused score is the maximum over branches, so that a strong indication from any branch is not diluted by the others,

$$S(x) = \max_m z_m(x) \quad (4)$$

and an alarm is raised when $S \geq \tau$, where τ is the 99th percentile of S on the validation set, i.e. a nominal 1% false-positive rate. No attack data are used at any point. A first pilot used the mean of empirical-CDF-transformed scores, but this diluted strong branches and was replaced by the maximum rule (see Section V-C).

D. Persistence Filter

Let $a_t \in \{0, 1\}$ be the per-window alarm. The filtered alarm a'_t is raised only if at least three of the last five per-window alarms are set, where $1[\cdot]$ is the indicator function:

$$a'_t = 1[a_t + a_{t-1} + a_{t-2} + a_{t-3} + a_{t-4} \geq 3] \quad (5)$$

E. Computational Footprint

The AE has 1,518 parameters, and the LOF reference set is $4,000 \times 22$ values (about 0.35 MB in single precision), so the whole detector fits comfortably on an industrial gateway.

Feature extraction needs only running sums over ten samples. Measured inference cost is reported in Section VI-C.

V. EXPERIMENTAL SETUP

A. Synthetic Substation Testbed

Because labelled attacks on real utility telemetry are not publicly available at sample-level resolution for this study, we built a synthetic substation simulator (supplementary code accompanies this paper). A per-unit load profile combines a compressed daily cycle with a slower harmonic and a bounded random walk. Line voltage is nominally 415 V and sags by 3% with load; current is 180 A times the load factor; power factor is about 0.92 with load dependence; active power follows the balanced three-phase relation of Eq. (1) plus 0.4% measurement noise; reactive power follows from the power factor; frequency is 50 Hz with small droop and 0.02 Hz noise; transformer temperature follows a first-order lag of the squared load; and the network descriptors are noisy constants near 100 packets/s, 64 bytes, and 10 ms jitter. We assume one sample per second when converting sample counts to time.

B. Attack Injection

Four attack classes are injected into the test stream as episodes lasting 60–300 samples, separated by 500–1,500 normal samples, until about 12% of samples are attacked (24–30 episodes per run). Table II lists the injection models. The FDI attack deliberately adjusts the power reading by only 60% of the current manipulation, imitating an attacker with imperfect knowledge of the physics.

TABLE II
INJECTED ATTACK CLASSES

Attack	Channels	Injection model
FDI	I, P	$I \times (1 + \delta), P \times (1 + 0.6\delta), \delta \sim U(0.10, 0.25)$
Replay / freeze	Electrical (7)	All electrical channels held at the last good sample
DoS flood	Network (2)	Packet rate $\times U(4, 10)$; jitter $\times U(0.2, 0.5)$
Ramp drift	V	Linear ramp of the voltage reading up to $+U(5\%, 10\%)$

C. Protocol, Baselines, and Metrics

Each run generates 40,000 normal training samples, 12,000 normal validation samples, and a 40,000-sample test stream with attacks, giving 39,991 test windows (window label = label of its last sample; 4,822–4,952 attack windows per run). Five independent runs (seeds 11, 22, 33, 44, 55) regenerate all data and retrain all models; $\pm$ denotes the sample standard deviation across seeds. For every model, the alarm threshold is the 99th percentile of its validation scores.

Baselines are isolation forest (IF, 200 trees) [20], one-class SVM (OCSVM, RBF kernel, $\nu = 0.01$, 4,000 training windows) [22], LOF, principal-component analysis with reconstruction error retaining 95% of the variance (PCA), and the AE, each evaluated with the raw features and with the physics residual (“+phys”). Hyper-parameters are library defaults except where stated. We report precision, recall, and F1 for the attack class, the false-positive rate (FPR), ROC-AUC, per-attack recall, false-alarm events per hour (contiguous alarm runs on normal windows), and detection delay (samples from episode onset to

the first alarm). Design choices (features, fusion rule, filter) were finalised on a separate development seed (7) after a pilot on one evaluation seed showed that mean-of-CDF fusion underperformed; all other choices were fixed beforehand, and no per-model tuning was done. All code uses Python and scikit-learn [33] on a single CPU core.

VI. RESULTS AND DISCUSSION

A. Overall Detection Performance

TABLE III
DETECTION PERFORMANCE ON THE TEST STREAM (MEAN OVER FIVE SEEDS)

Model	Prec.	Recall	F1 ($\pm \sigma$)	FPR (%)	ROC-AUC
IF	0.449	0.293	0.281 ± 0.106	9.7	0.743
OCSVM	0.734	0.654	0.684 ± 0.088	3.8	0.852
LOF	0.783	0.672	0.720 ± 0.080	2.7	0.867
PCA	0.775	0.664	0.710 ± 0.096	3.1	0.882
AE	0.821	0.658	0.729 ± 0.073	2.0	0.888
IF+phys	0.480	0.369	0.336 ± 0.127	10.7	0.828
OCSVM+phys	0.796	0.965	0.868 ± 0.076	3.8	0.983
LOF+phys	0.843	0.966	0.899 ± 0.039	2.6	0.983
PCA+phys	0.797	0.698	0.740 ± 0.076	2.7	0.920
AE+phys	0.875	0.955	0.913 ± 0.014	1.9	0.980
Fusion	0.860	0.964	0.908 ± 0.023	2.2	0.984
(AE+LOF+phys)					
Fusion + 3-of-5	0.891	0.952	0.920 ± 0.019	1.6	0.984
AE+phys + 3-of-5	0.907	0.943	0.925 ± 0.014	1.3	0.980

Rows without “+phys” use only the ten raw channels; “+phys” adds the residual of Eq. (1). The nominal FPR is 1%. σ is the sample standard deviation over five seeds.

Table III summarises the results, Fig. 2 illustrates one test segment, and Fig. 3 shows ROC curves. Without the physics residual, no detector exceeds an F1-score of 0.729 (AE) and ROC-AUC of 0.888. With the residual, OCSVM, LOF, and AE reach F1-scores of 0.868, 0.899, and 0.913 and ROC-AUC of about 0.98. The fused detector attains the (marginally) highest ROC-AUC (0.984) and, with the persistence filter, an F1-score of 0.920 ± 0.019 at 1.6% FPR. Isolation forest performs worst in every setting ($F1 \leq 0.34$), and PCA benefits little from the residual, although we did not investigate why.

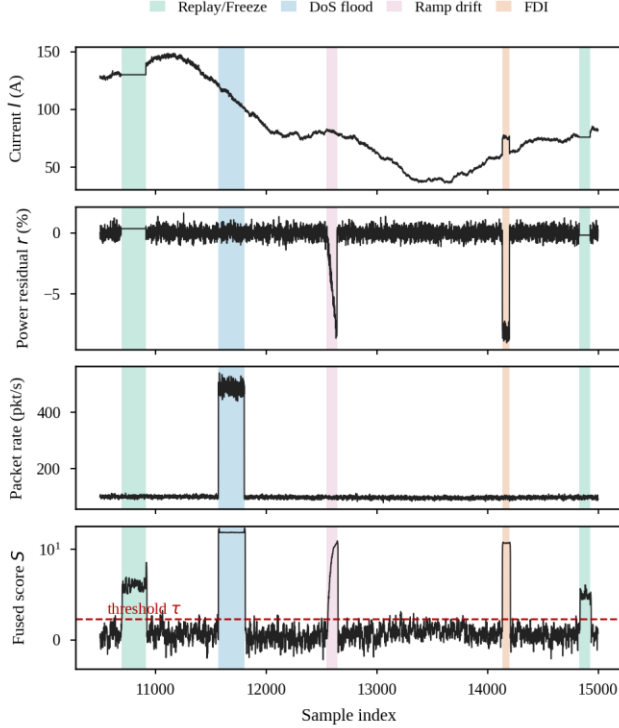


Fig. 2. One test segment (seed 11). Shaded bands mark attack episodes. The current alone shows little that distinguishes FDI from a load change, whereas the residual, packet rate, and fused score expose all four attack classes.

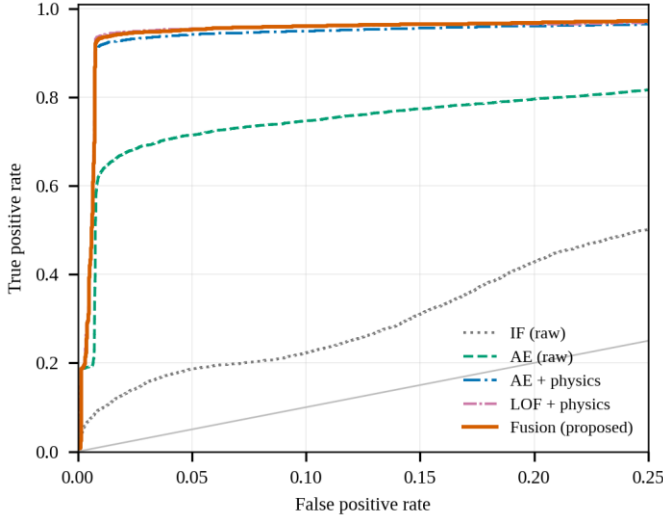


Fig. 3. ROC curves (seed 11, false-positive rate up to 0.25). The physics residual shifts the AE curve upward; fusion tracks the best single branch.

B. Effect of the Physics Residual and Per-Attack Behaviour

Figs. 3 and 4 show where the improvement comes from. For FDI, the AE recall rises from 10.6% to 100.0%: the manipulated current and power stay inside their normal ranges, so a detector using only raw statistics has little to detect, whereas the residual shifts by roughly 4–9% (Fig. 2), far above its noise floor of well under 1%. Ramp-drift recall increases from 71.7% to 87.9% for the same reason. DoS flooding and replay/freeze are already detected from network and variance statistics (recall $\geq 96\%$) and are unaffected. The choice of model matters less than the choice of feature: with the residual, OCSVM, LOF, and AE all

detect essentially every FDI window, whereas IF (21%) and PCA (26%) do not.

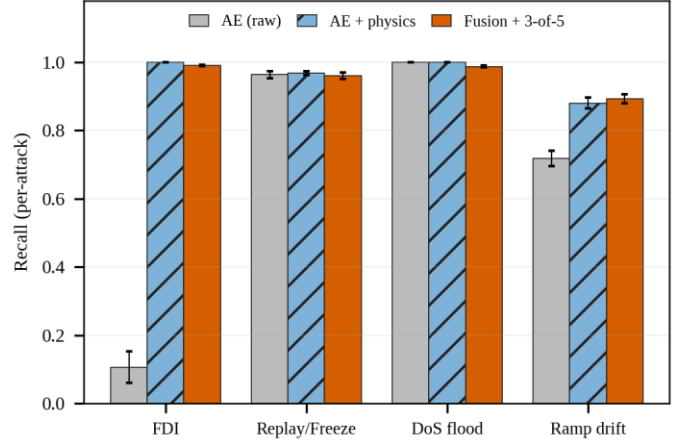


Fig. 4. Per-attack recall (mean and standard deviation over five seeds) for the raw autoencoder, the autoencoder with the physics residual, and the fused detector with persistence filter.

C. Detection Delay and Computational Cost

TABLE IV
DETECTION DELAY OF THE FUSED DETECTOR WITH PERSISTENCE FILTER

Attack	Episodes	Median	Mean	Max
FDI	36	2.0	1.9	2.0
Replay / freeze	25	7.0	6.6	9.0
DoS flood	34	2.0	2.0	2.0
Ramp drift	44	18.0	19.3	43.0

Delay is measured in samples (about seconds at 1 Hz) from episode onset to the first alarm; pooled over five seeds, with no missed episodes.

Table IV shows that abrupt attacks are flagged within two samples, replay/freeze within about seven (the window variance must collapse first), and the slow ramp drift after a median of 18.0 samples, with a worst case of 43.0. No episode was missed. On a single CPU core, batch inference costs about 0.0006 ms per window for the AE and 0.0237 ms for LOF, and the AE trains in about 8.5 s. These numbers come from a general-purpose processor and will be higher on embedded gateways, but they leave ample headroom at one-second sampling.

D. Fusion, Persistence, and Alert Burden

TABLE V
EFFECT OF SCORE FUSION AND THE 3-OF-5 PERSISTENCE FILTER

Configuration	F1	FPR (%)	ROC-AUC	False-alarm events / h
AE+phys	0.913	1.9	0.980	31.2 $\pm$ 7.6
AE+phys + 3-of-5	0.925	1.3	0.980	7.1 $\pm$ 1.7
Fusion (AE+LOF+phys)	0.908	2.2	0.984	34.1 $\pm$ 13.0
Fusion + 3-of-5	0.920	1.6	0.984	9.1 $\pm$ 4.3

False-alarm events are contiguous alarm runs on attack-free windows per 3,600 samples (one hour at 1 Hz).

Table V separates the contributions of fusion and persistence. The persistence filter is effective: it cuts false-alarm events from about 34.1 to 9.1 per hour for the fused detector, at a recall cost of about one percentage point. However, score fusion itself brings no significant benefit in this testbed. The AE with the residual and the 3-of-5 filter is at least

as good ($F1\ 0.925 \pm 0.014$, 7.1 false-alarm events per hour), and the ROC-AUC difference to the fused score (0.984 versus 0.980) is within one standard deviation. We therefore regard fusion as an optional robustness measure for deployments where attack types are less predictable, not as a demonstrated accuracy gain, and recommend the simpler AE-plus-residual configuration when the attack classes resemble those tested here.

E. Threats to Validity

Four caveats apply. First, the data are synthetic; absolute numbers will not transfer to real substations, whose noise, load dynamics, and protocol behaviour are richer, and the attacks are simple parametric models. Second, realised FPRs are above the nominal 1% (1.9–3.8% for the useful detectors, roughly 10% for IF), which indicates a distribution shift between the validation and test periods and argues for periodic recalibration. Third, even after filtering, 9.1 false-alarm events per hour at 1 Hz is too many for direct operator paging; in practice alerts would be aggregated, ranked, or gated by a higher threshold, at some cost in delay and recall. Fourth, the fusion rule was revised once after a pilot on an evaluation seed, and the development seed and the five evaluation seeds share the same simulator, so the estimates may be somewhat optimistic.

VII. LIMITATIONS AND FUTURE WORK

Stealthy attackers. The physics residual is only as strong as the invariant. An attacker who alters voltage, current, power factor, and power consistently, as in the undetectable attacks of Liu *et al.* [14], preserves the residual and would be missed by that branch; countering this requires redundancy across independent measurements (for example, neighbouring meters, phasor data, or state estimation) and authenticated sensing. **Real data.** We plan to evaluate on public testbed data such as SWaT [26] and on utility traces, and to add sequence models such as LSTM [24], [25] as additional branches. **Drift and lifecycle.** Non-stationary load requires scheduled recalibration and controlled retraining through the model registry of Fig. 1. **Explainability and privacy.** Feature-attribution methods such as SHAP [34] could tell operators which channel drove an alarm, and federated learning [35] could train across substations without centralising raw telemetry. **Adversarial robustness.** Evasion attacks that adapt to the detector remain outside the scope of this study and should be evaluated before deployment in safety-critical settings.

VIII. CONCLUSION

This paper presented a secure IIoT architecture and an unsupervised, edge-deployable anomaly detector for smart-grid telemetry that exploits a cheap physical invariant. In a controlled synthetic study, adding the power-balance residual raised FDI recall for an autoencoder from 11% to 100% and lifted F1 from 0.729 to 0.913, a larger gain than any change of learning model, while a 3-of-5 persistence filter cut false-alarm events by roughly three quarters. Multi-detector score fusion did not add a significant benefit. The results are limited by the synthetic data and by the assumption of a non-adaptive attacker,

and validation on real utility data is the most important next step.

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